Activity: Logistic regression

Data

Researchers collected data on 147 individuals in the Netherlands, 67 of whom kept a bird. The data include the following variables:

- `Cancer`: whether the individual developed lung cancer (0 = no, 1 = yes)
- `Bird`: whether the individual kept a bird at home for at least 6 consecutive months from 5 to 14 years before diagnosis (cases) or examination (controls)
- `Sex`: male or female
- `SocEc`: a rough measure of socioeconomic status. `Low` or `High`, depending on the principal wage earner in the household
- `Age`: age in years
- `SmokeYrs`: years of smoking prior to diagnosis or examination
- `SmokeRate`: average number of cigarettes smoked per day

Logistic regression model

Using the data, the researchers fit a logistic regression model:

```r
m_bird <- glm(Cancer ~ Bird + Sex + SocEc + Age +
                SmokeYrs + SmokeRate,
              data = birdkp, family = binomial)

coef(m_bird)
```

```
(Intercept)        Bird     SexMale    SocEcLow         Age    SmokeYrs 
-1.27063830  1.36259456 -0.56127270 -0.10544761 -0.03975542  0.07286848 
  SmokeRate 
 0.02601689 
```

Questions

1. Interpret the estimated coefficient 1.363 in context.

2. The estimated coefficient on Age is negative. Why might that be the case? Is there anything you would change in the model?

3. What is the predicted probability of lung cancer for a male individual who keeps a bird, of “high” socioeconomic status, aged 35, who has never smoked?